

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE BIOLOGY

Paper 1

Monday 22 May 2023

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

This question is about photosynthesis.

0 1 . 1

Which plant organ absorbs water from the soil for photosynthesis?

[1 mark]

Tick (✓) **one** box.

Leaf

☐

Root

☐

Stem

☐

0 1 . 2

Which part of a plant cell absorbs light for photosynthesis?

[1 mark]

Tick (✓) **one** box.

Cell wall

☐

Chloroplast

☐

Nucleus

☐

Vacuole

☐

0 1 . 3

Carbon dioxide is needed by a plant for photosynthesis.

How does carbon dioxide enter a plant?

[1 mark]

Tick (✓) **one** box.

By active transport

☐

By diffusion

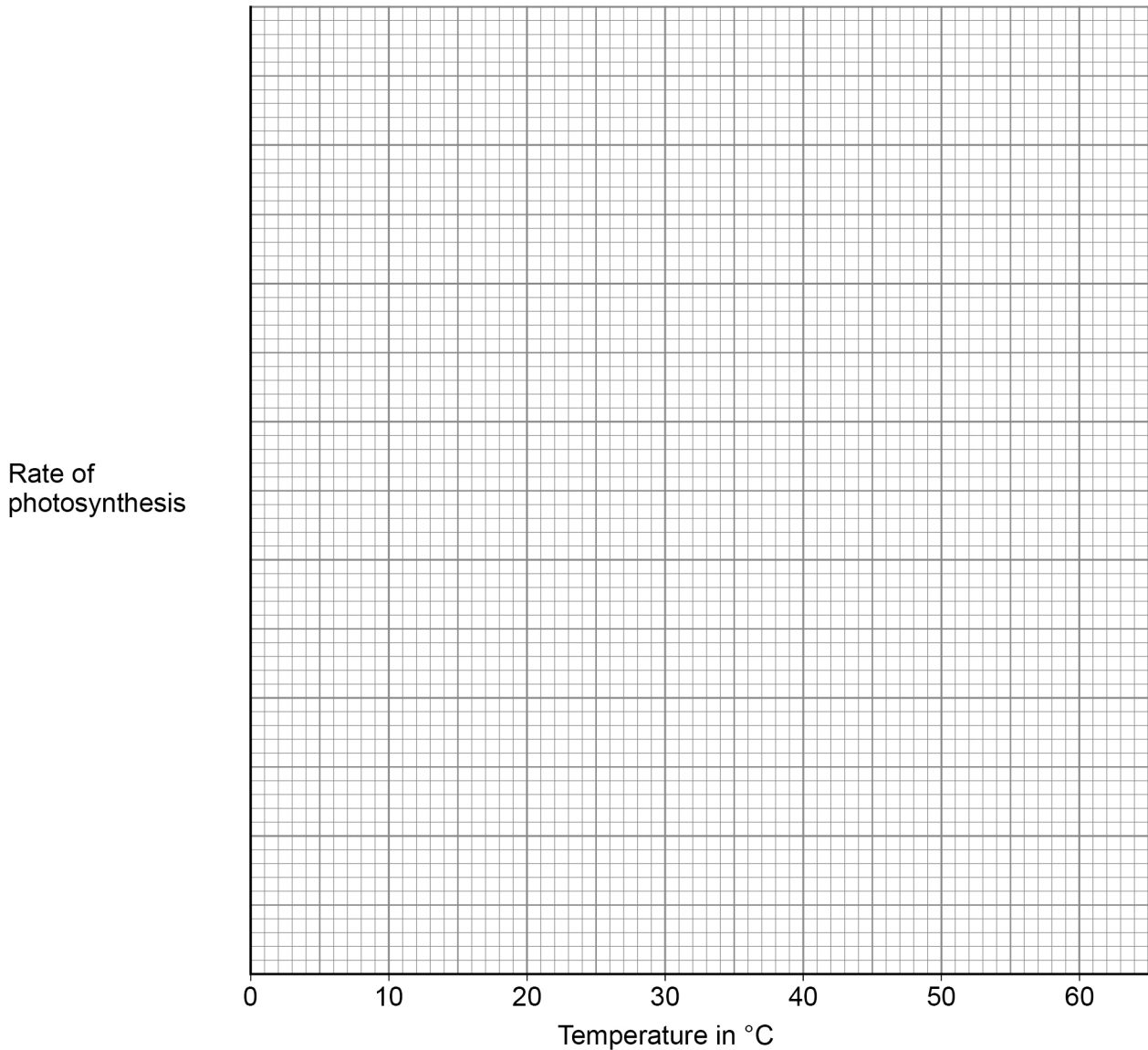
☐

By osmosis

☐

The rate of photosynthesis is affected by temperature.

Figure 1



0 1 . 4

Sketch a line on **Figure 1** to show the relationship between temperature and the rate of photosynthesis.

[2 marks]

Question 1 continues on the next page

Turn over ►



0 1 . 5 One product of photosynthesis is glucose.

Glucose is a sugar.

Which tissue transports dissolved sugars through a plant?

[1 mark]

Tick (✓) **one** box.

Epidermis

☐

Palisade mesophyll

☐

Phloem

☐

0 1 . 6 Give **two** uses of glucose in a plant.

[2 marks]

1

2

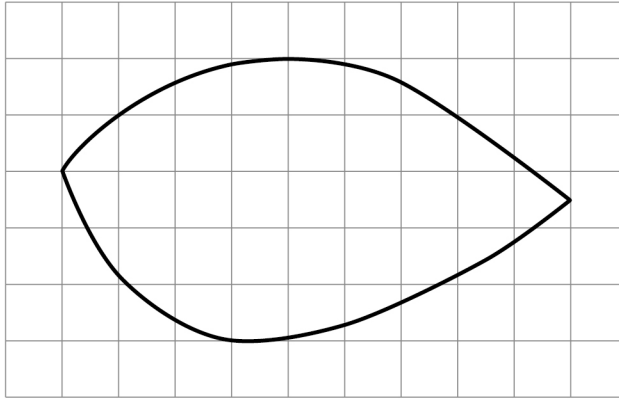


A student used a leaf in an investigation.

The student drew an outline of the leaf on squared paper.

Figure 2 shows the outline.

Figure 2



 = 1 cm²

0 1 . 7 Estimate the surface area of the leaf in **Figure 2**.

[1 mark]

Surface area = _____ cm²

Question 1 continues on the next page

Turn over ►



0 1 . 8 The student used a different leaf.

The area of the leaf was 22 cm^2 .

The leaf released 0.04 cm^3 of oxygen per cm^2 of leaf every hour.

Calculate the volume of oxygen released by the leaf in **two** hours.

[3 marks]

Volume = _____ cm^3

12



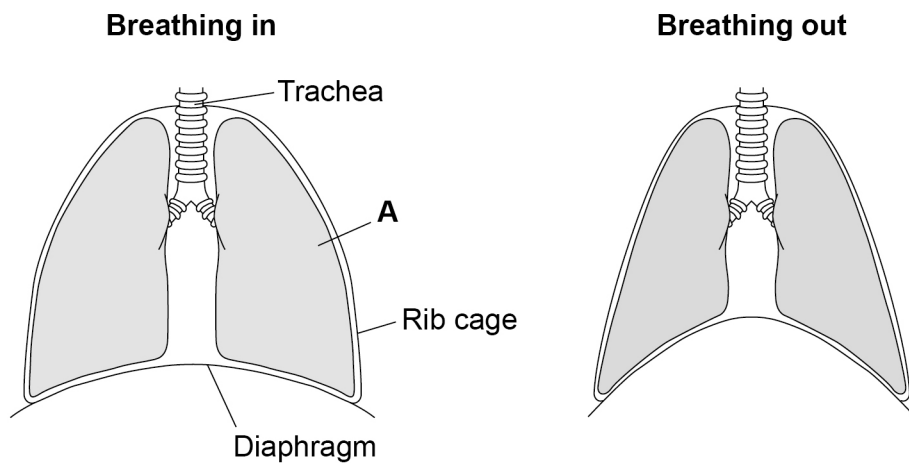
0 2

The human breathing system takes air into and out of the body.

Figure 3 shows part of the breathing system:

- when breathing in
- when breathing out.

Figure 3



0 2 . 1

Name part **A** on **Figure 3**.

[1 mark]

0 2 . 2

Describe **two** changes that take place in the breathing system when breathing in.

[2 marks]

1

2

Question 2 continues on the next page

Turn over ►



A student investigated the effect of exercise on breathing rate.

This is the method used.

1. Count the number of breaths at rest.
2. Carry out an exercise.
3. Count the number of breaths after exercise.
4. Calculate the change in the number of breaths.

0 2 . 3 The student needs to count the number of breaths in a given time.

What time would give the most accurate results?

[1 mark]

Tick (✓) **one** box.

5 seconds

☐

1 minute

☐

1 hour

☐

0 2 . 4 What equipment should the student use to measure the time in this investigation?

[1 mark]

0 2 . 5 The method instructs the student to 'carry out an exercise'.

Suggest **two** ways this instruction could be improved.

[2 marks]

1

2



0 2 . 6 Many factors damage the breathing system.

Give **one** factor that may damage the breathing system.

[1 mark]

0 2 . 7 A healthy person continually breathes efficiently.

Name **one** piece of equipment that can help a patient to breathe if they stop breathing efficiently.

[1 mark]

9

Turn over for the next question

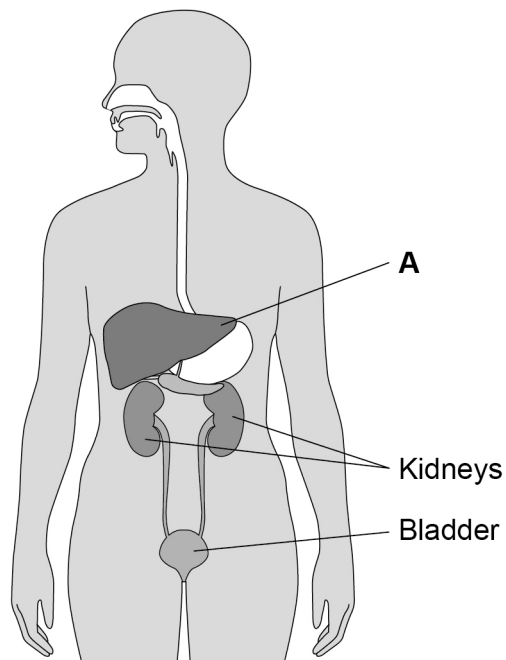
Turn over ►



0 3

Figure 4 shows organs involved in controlling conditions in the body.

Figure 4



0 3

1

Name organ **A** on **Figure 4**.

[1 mark]

0 3

2

The water content of the body is automatically controlled.

Give **two** other conditions that are automatically controlled in the body.

Do **not** refer to water in your answer.

[2 marks]

0 3

3

Some water leaves the body through the skin.

Name the process by which water leaves the body through the skin.

[1 mark]



The kidney filters blood to remove waste products.

The liquid that passes from the blood into the kidney contains a mixture of dissolved substances. The liquid is called kidney filtrate.

Table 1 shows the concentration of some substances in the blood, kidney filtrate and urine.

Table 1

Substance	Concentration in g per dm ³		
	Blood	Kidney filtrate	Urine
Protein	70	0	0
Glucose	1	1	0
Urea	0.3	0.3	20
Sodium ions	3	3	6

0 3 . 4

Name the substance in **Table 1** that is **not** filtered out of the blood by the kidney.

[1 mark]

0 3 . 5

Name the substance in **Table 1** that is completely reabsorbed into the blood.

Give evidence from **Table 1** for your answer.

[2 marks]

Substance _____

Evidence _____

Question 3 continues on the next page

Turn over ►



0 3 . 6 A person who has kidney failure can be treated by:

- dialysis
- kidney transplant.

In dialysis the patient's blood is passed through filters to remove waste products.

Table 2 gives some information about the two treatments.

Table 2

Dialysis	Kidney transplant
A tube is attached to a needle in the person's arm. Blood passes out through the tube and into a machine that filters the blood before it is returned to the body.	Complicated surgery that takes several hours. Risks of surgery include blood clots forming.
Dialysis can be used for 5 to 10 years before a patient must have a transplant.	A kidney transplant lasts 15 to 20 years.
Usually carried out at a dialysis centre three times a week. It takes 4 hours on each occasion to complete.	Hospital check-ups every 3 months are required for the rest of the person's life.
Dialysis is available immediately at most large hospitals.	On average it takes 2 to 3 years for a suitable donor kidney to become available.
Restricted diet and fluid intake needed to maintain good health.	No specific diet restrictions.



Use information from **Table 2**.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

13

Turn over ►



0 4

The heart pumps blood around the body.

0 4 . 1

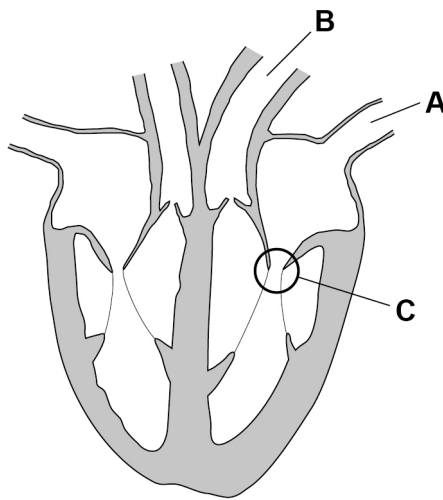
The natural resting heart rate is controlled by a group of cells.

What is the name of this group of cells?

[1 mark]

Figure 5 shows the heart.

Figure 5



0 4 . 2

What is the name **and** function of part C?

[2 marks]

Name _____

Function _____



[illegible]

Turn over ►



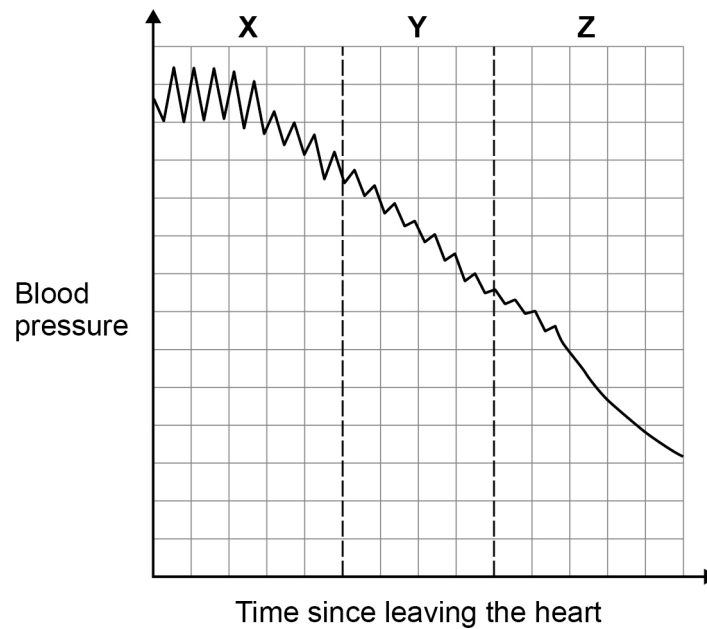
0 4 . 4

Blood pressure is the force of blood on the walls of blood vessels.

Blood passes through three types of vessels, **X**, **Y** and **Z**.

Figure 6 shows changes in the blood pressure after it leaves the heart.

Figure 6



Which type of blood vessel is **X**?

Give a reason for your answer.

[2 marks]

Type of blood vessel _____

Reason _____

10



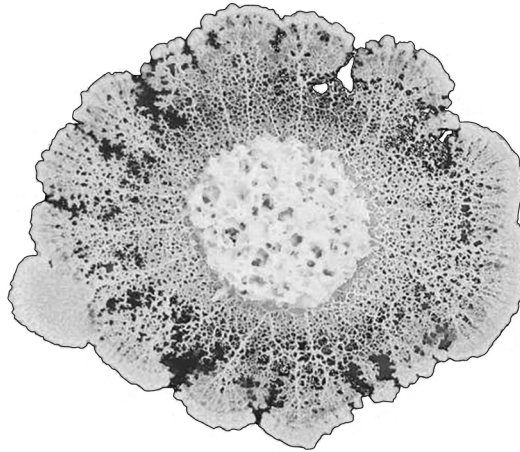
0 5

A slime mould is a simple organism.

A slime mould can be single-celled or multicellular.

Figure 7 shows a slime mould.

Figure 7

**0 5****. 1**

A circular slime mould had a radius of 13 mm.

Calculate the perimeter of the slime mould in centimetres.

Use the equation:

$$\text{perimeter} = 2 \times 3.14 \times \text{radius}$$

Give your answer in cm.

[3 marks]

Perimeter = _____ cm

Question 5 continues on the next page

Turn over ►



0 5 . 2 A multicellular slime mould does **not** have any organ systems.

A slime mould does **not** have a stomach.

Give **three** other organs found in the human digestive system that would **not** be found in the slime mould.

[3 marks]

1 _____

2 _____

3 _____

0 5 . 3 Explain how the slime mould uses phagocytosis and enzymes to obtain nutrients from particles of food.

[3 marks]



Question 5 continues on the next page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



As the slime mould moves it leaves a trail of sticky liquid.

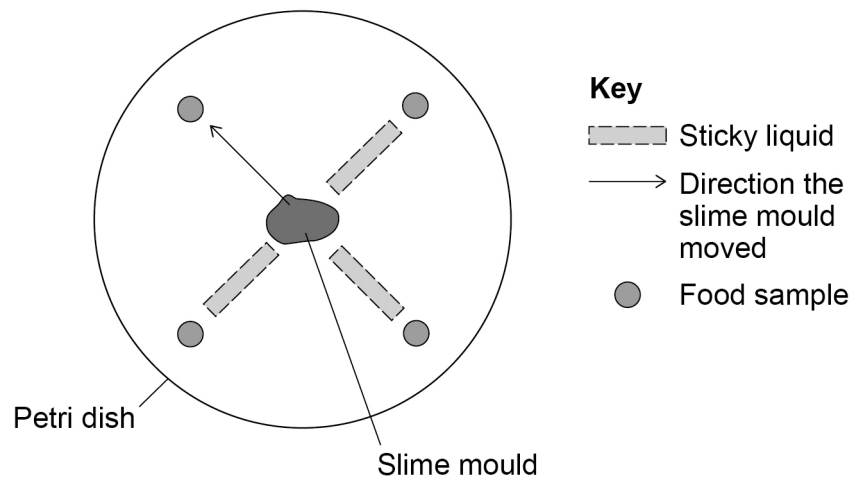
Scientists investigated the effect of the sticky liquid on the movement of the slime mould.

This is the method used.

1. Place four samples of the same food on a Petri dish.
2. Add three trails of the sticky liquid to the Petri dish.
3. Place a slime mould in the centre of the Petri dish.
4. Observe how the slime mould moves.

Figure 8 shows the Petri dish and the results of the investigation.

Figure 8



0 5 . 4

What conclusion can the scientists make about the movement of the slime mould in **Figure 8**?

[1 mark]

0 5 . 5

Suggest how the movement shown in **Figure 8** could benefit the slime mould in the conditions where it normally lives.

[1 mark]



0 5 . 6

Slime moulds can detect light.

Suggest how slime moulds detect light.

[1 mark]

0 5 . 7

Slime moulds commonly live in areas of shade.

Suggest why living in shade is an advantage to a slime mould.

[1 mark]

A single slime mould can reproduce by forming spore cells.

Spore cells have half the number of chromosomes compared to the slime mould.

0 5 . 8

Name the type of cell division that forms the spore cells.

[1 mark]

0 5 . 9

Two spore cells fuse to form a zygote.

Explain why the zygote is **not** genetically identical to the original slime mould cells.

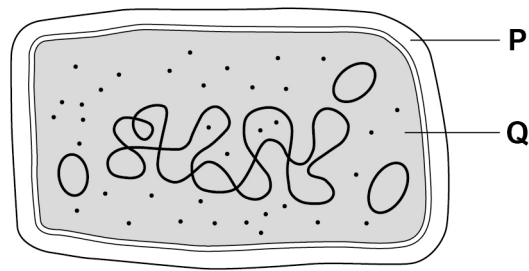
[3 marks]



0 6

Figure 9 shows a bacterial cell.

Figure 9



0 6

1

Name parts **P** and **Q** on **Figure 9**.

[2 marks]

P _____

Q _____

0 6

2

Bacterial cells and animal cells both contain genes.

Give **two** differences in the location of genes in a bacterial cell compared to in an animal cell.

[2 marks]

1 _____

2 _____

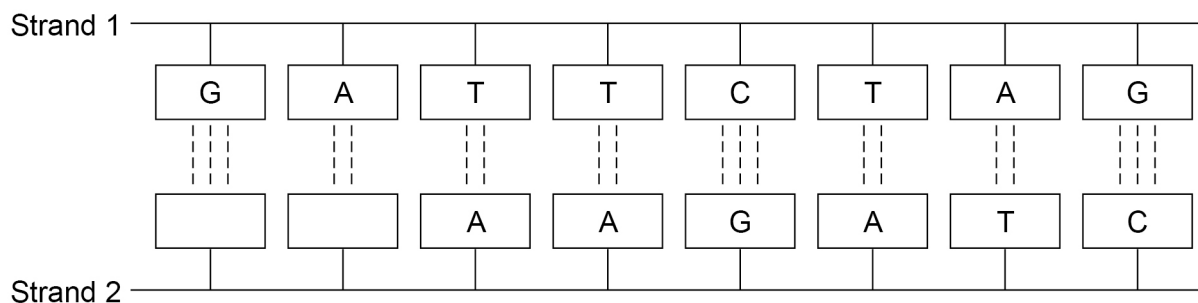


Genes are sections of DNA.

DNA contains combinations of the four bases, **A**, **C**, **G** and **T**.

Figure 10 shows the bases in a section of a double strand of DNA.

Figure 10



Key

---- Bonds between DNA bases

0 6 . 3 Complete **Figure 10** by writing **A**, **C**, **G** or **T** in each of the **two** empty boxes.

[1 mark]

0 6 . 4 The two strands of DNA form a twisted spiral structure.

What is the scientific term for this structure?

[1 mark]

Question 6 continues on the next page

Turn over ►



0	6	.	5
---	---	---	---

Cystic fibrosis is an inherited condition.

The allele for cystic fibrosis is recessive.

A man and a woman, both without cystic fibrosis, have a child with the condition.

What is the probability that the parents' next child will have cystic fibrosis?

You should:

- draw a genetic diagram
- identify all the offspring with cystic fibrosis
- use the following symbols:

A = allele for not having cystic fibrosis

a = allele for cystic fibrosis.

[5 marks]

Probability of cystic fibrosis = _____



0	6	.	6
---	---	---	---

Cystic fibrosis is caused when a sequence of three bases in a gene is changed.

Explain how a change in the sequence of three bases could lead to a non-functioning protein.

[3 marks]

Question 6 continues on the next page

Turn over ►



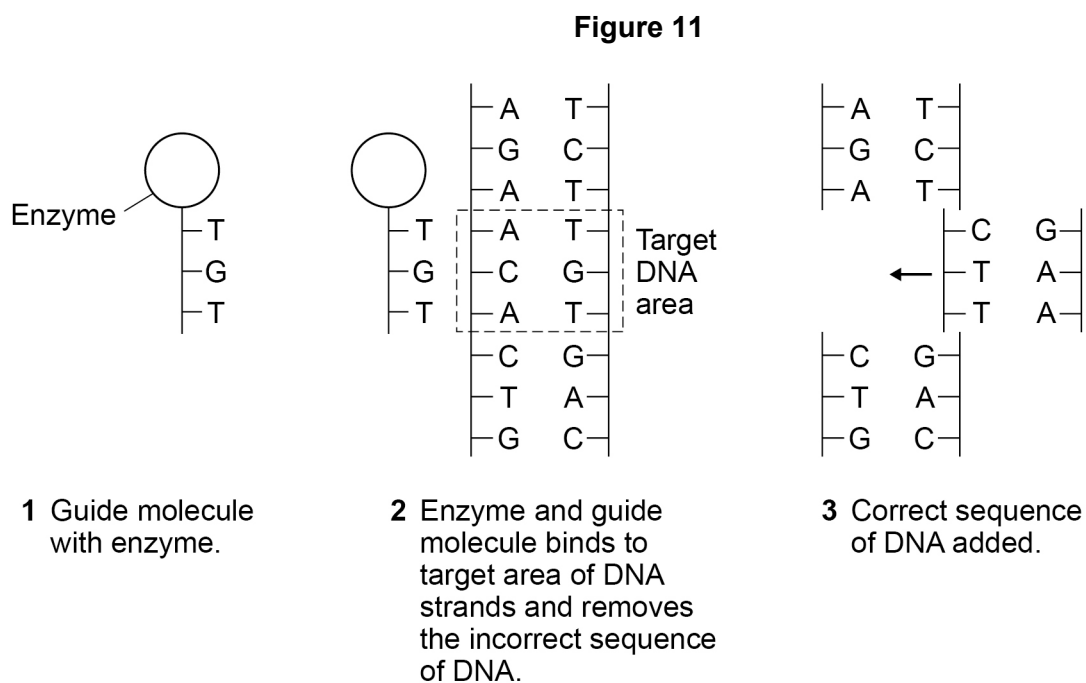
Scientists have found an enzyme in bacteria which cuts DNA.

The enzyme can be used in a new technique called CRISPR to change genes.

These are the steps in the CRISPR technique.

1. A guide molecule which binds to a specific DNA sequence is made.
2. The enzyme is attached to the guide molecule.
3. The guide molecule with the enzyme binds to the DNA strands.
4. The enzyme cuts the DNA strands.
5. The faulty sequence of bases is removed and the correct sequence is added.

Figure 11 shows the CRISPR technique.



0 6 . 7

Cystic fibrosis is one condition scientists are hoping to treat with CRISPR.

Describe how scientists could use the CRISPR technique to treat cystic fibrosis.

[3 marks]

0 6 . 8

Changing a person's genes is illegal in many countries.

Suggest **one** reason why.

[1 mark]

18

Turn over for the next question

Turn over ►



0 7

The orangutan has been declared a critically endangered species.

0 7 . 1

There are estimated to be about 95 600 orangutans living in the wild.

The number of orangutans is predicted to decrease by 15% by 2026.

Calculate the predicted number of orangutans in 2026.

[3 marks]

Predicted number of orangutans = _____

0 7 . 2

A rehabilitation centre for orangutans was set up in 1964.

The rehabilitation centre teaches the orangutans survival skills including nest building and climbing skills.

When the orangutans are fully rehabilitated they are released into a wildlife reserve.

Suggest why nest building and climbing skills are important for survival in the wild.

[2 marks]



0 7 . 3

Mating behaviour has been studied in adult orangutans in the wild.

Both younger and older males have been found to mate and produce offspring at similar rates.

- When male orangutans reach an older age they double in mass and develop secondary sexual characteristics:
 - a flanged face
 - a large throat sac.

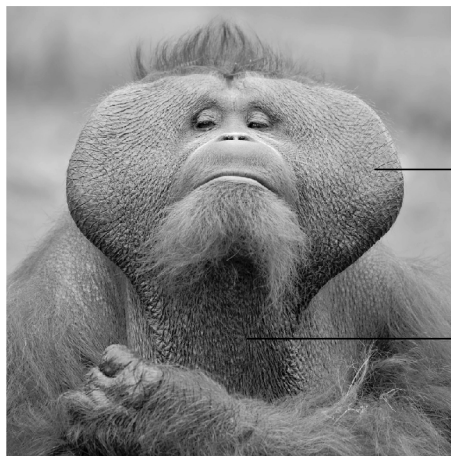
Figure 12 shows the differences in appearance of younger male and older male orangutans.

Figure 12

Young male orangutan



Old male orangutan



Flange

Throat sac

Younger males travel large distances compared to older males. Older males spend a lot of their time resting and feeding.

Older males are able to use their large throat sacs to make loud calls across long distances.

Suggest how younger male orangutans **and** older male orangutans are able to successfully find mates.

[3 marks]

Turn over ►



0	7	.	4
---	---	---	---

Both younger males and older males eat large amounts of fruit as their main food source.

Fruit is high in sugars.

Explain why older males have a greater mass compared to the younger males.

[3 marks]

11

END OF QUESTIONS



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



*Do not write
outside the
box*

[illegible]

[illegible]

*Do not write
outside the
box*

[illegible]

Do not write
outside the
box

[illegible]

There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Copyright information

For confidentiality purposes, all acknowledgements of third-party copyright material are published in a separate booklet. This booklet is published after each live examination series and is available for free download from www.oxfordaqaexams.org.uk.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and Oxford International AQA Examinations will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team.

Copyright © 2023 Oxford International AQA Examinations and its licensors. All rights reserved.



3 6



2 3 6 Y 9 2 0 1 / 1

IB/M/Jun23/9201/1